Evaluating the Performance of Large Language Models in Drafting Data Management Plans

Nahid Zeinali[1], Xuebin Dong[1], Rebecca Hofstein Grady[2], Maria Praetzellis[2], Brian Riley[2], Bhavesh Patel[1]

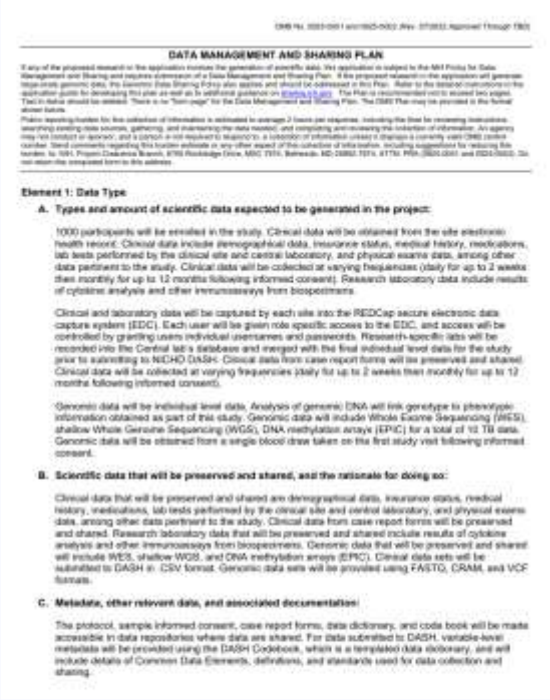
[1] FAIR Data Innovations Hub, California Medical Innovations Institute, San Diego, California, United States of America, 92121
[2] California Digital Library, University of California Office of the President, Oakland, CA, United States of America, 94607

What Are Data Management Plans?

- Most funders require Data Management Plans **(DMPs)** as part of a grant proposal.

- A DMP is a document that describes how researchers will manage, preserve, and share data from their projects in line with the **FAIR Principles**.
  NIH expanded its DMP requirement in **January 2023.**
  NSF has required DMPs for proposals since **January 2011.**

- **DMP Tool** is a free, community-supported service that makes it easier to create machine-actionable data management and sharing plans (DMSPs) that meet funder requirements and follow open science best practice.



DATA MANAGEMENT AND SHARING PLAN

Element 1: Data Type

A. Types and amount of scientific data expected to be generated in the project:

1000 participants will be enrolled in the study. Clinical data will be obtained from the site electronic health record. Clinical data include demographical data, insurance status, medical history, medications, lab tests performed by the clinical site and central laboratory, and physical exams data, among other data pertinent to the study. Clinical data will be collected at varying frequencies (daily for up to 2 weeks then monthly for up to 12 months following informed consent). Research laboratory data include results of cytokine analysis and other immunoassays from biospecimens.

Clinical and laboratory data will be captured by each site into the REDCap secure electronic data capture system (EDC). Each user will be given role specific access to the EDC, and access will be controlled by granting users individual usernames and passwords. Research-specific labs will be recorded into the Central lab's database and merged with the final individual level data for the study prior to submitting to NICHD DASH. Clinical data from case report forms will be preserved and shared. Clinical data will be collected at varying frequencies (daily for up to 2 weeks then monthly for up to 12 months following informed consent).

Genomic data will be individual level data. Analysis of genomic DNA will link genotype to phenotypic information obtained as part of this study. Genomic data will include Whole Exome Sequencing (WES), shallow Whole Genome Sequencing (WGS), DNA methylation arrays (EPIC) for a total of 10 TB data. Genomic data will be obtained from a single blood draw taken on the first study visit following informed consent.

B. Scientific data that will be preserved and shared, and the rationale for doing so:

Clinical data that will be preserved and shared are demographical data, insurance status, medical history, medications, lab tests performed by the clinical site and central laboratory, and physical exams data, among other data pertinent to the study. Clinical data from case report forms will be preserved and shared. Research laboratory data that will be preserved and shared include results of cytokine analysis and other immunoassays from biospecimens. Genomic data that will be preserved and shared will include WES, shallow WGS, and DNA methylation arrays (EPIC). Clinical data sets will be submitted to DASH in .CSV format. Genomic data sets will be provided using FASTQ, CRAM, and VCF formats.

C. Metadata, other relevant data, and associated documentation:

The protocol, sample informed consent, case report forms, data dictionary, and code book will be made accessible in data repositories where data are shared. For data submitted to DASH, variable-level metadata will be provided using the DASH Codebook, which is a templated data dictionary, and will include details of Common Data Elements, definitions, and standards used for data collection and sharing.

Example of an NIH DMP

https://dmptool.org/

Challenges in Writing DMP

Knowledge & Training

Researchers often lack training in FAIR data practices.

Data management and sharing requirements can be difficult to understand.

Funder-Specific Requirements

DMPs differ substantially across funding agencies.

Terminology, formats, and requirements vary from one funders to another.

Limited Support Resources

Researchers at smaller and independent institutions may lack librarian support.

Librarians may have varying levels of FAIR and funder-specific expertise.

https://www.sciencedirect.com/science/article/pii/S1532046417300990

Motivation and Study Objective

Large Language Models

Large Language Models (LLMs) have demonstrated potential in technical writing.

- Patent abstract generation
- Clinical trial protocol drafting
- Complex technical writing tasks

Purpose

Evaluate the performance of Large Language Models (LLMs) in drafting Data Management Plans (DMPs) compliant with the NIH 2023 Data Management and Sharing Policy.

https://arxiv.org/abs/2507.22387;https://arxiv.org/abs/2404.05044

Study Design and Evaluation Framework

DMP Generation

Two LLMs compared:

- **GPT-4.1** (OpenAI, commercial)
- **Llama 3.3 70B** (Meta, open-source)

Prompting strategy (minimal prompt):

- NIH Institute / Center name
- Data type, subject type & consent details
- Full NIH DMP template

10 runs per sample per model

Evaluation Framework

Benchmark: All 26 NIH DMP samples used as reference

1- Automatic Reference-Based Evaluation

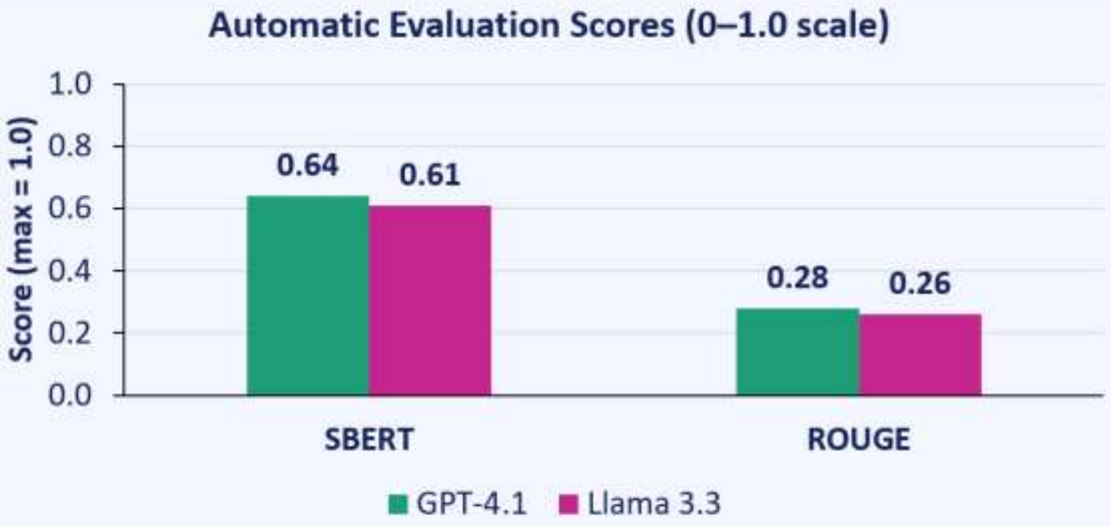
- Evaluated Metrics **(SBERT and ROUGE)** at 3 levels of granularity

2- Human Expert Evaluation

Reviewers blindly evaluated 3 different DMP: (One from original NIH sample DMP, one Llama 3.3-generated DMP, one GPT-4.1-generated DMP)

- **Satisfaction** (1–5 scale)
- **Error types** (accuracy, completeness, compliance, etc.)
- **Human vs. AI** detection judgment

Results: Automatic Evaluation (n=26 DMPs)



The automatic reference-based evaluation scores for both **ROUGE** and **SBERT** were slightly higher for GPT than for Llama

Human Evaluation Result: Satisfaction

Satisfied or Very Satisfied (%)

90.5%	42.9%	19.0%
GPT-4.1	**Human**	**Llama 3.3**

On a 5-point Likert satisfaction scale, experts reported higher satisfaction with **GPT-generated DMPs** compared to Llama and even human-generated ones.

Error Analysis and AI Detectability

Average Errors per DMP by Model

2.3
GPT-4.1

7.0
Human

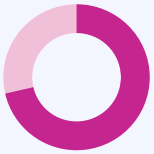
7.3
Llama 3.3

Overall, GPT showed the lowest error rates across error types in the assessment, but some clarity and completeness errors were still pointed out by participants

Human vs. AI Detection (% correct)

14.3%
GPT-4.1
correctly ID'd as AI

66.7%
Human
correctly ID'd as human



71.4%
Llama 3.3
correctly ID'd as AI

GPT outputs mistaken for human-written 85.7% of the time

Discussion, Conclusion, and Future Direction

What we found

- Built a template-agnostic evaluation framework and benchmark for LLM-generated DMPs.
- GPT slightly outperformed Llama in automated evaluation but showed higher performance in human assessment.
- Open-source LLMs may not be as ready off-the-shelf as commercial ones for drafting funding-compliant DMPs.
- GPT-generated DMPs still contained errors, highlighting the need for expert review to ensure accuracy, completeness, and compliance.

Future work

- Extend the evaluation to more funders such as (NSF) and the updated NIH guidelines
- Enhance open-source LLMs using RAG for improved DMP drafting.
- Develop tools for automatic machine-actionable DMP generation.
- Support consistent LLM-based DMP generation and evaluation across funders.

Thank You!

Questions? Reach out to us

Nahid Zeinali: nzeinali@calmi2.org

Find these slides and all resources here